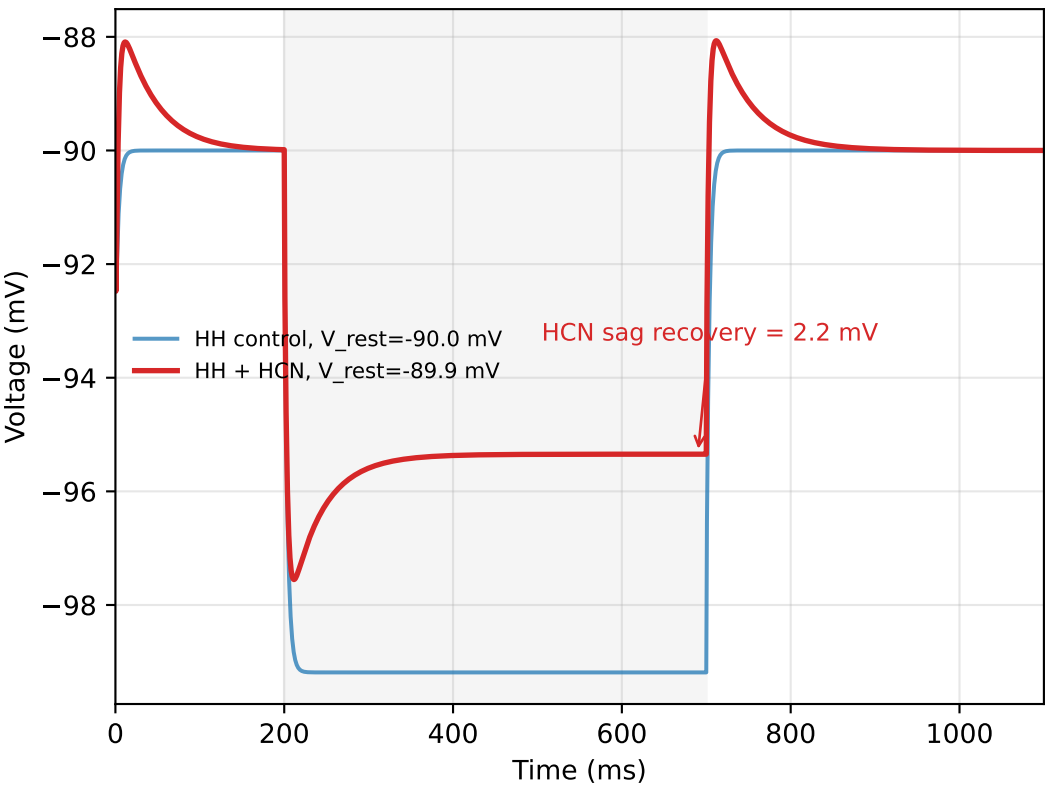
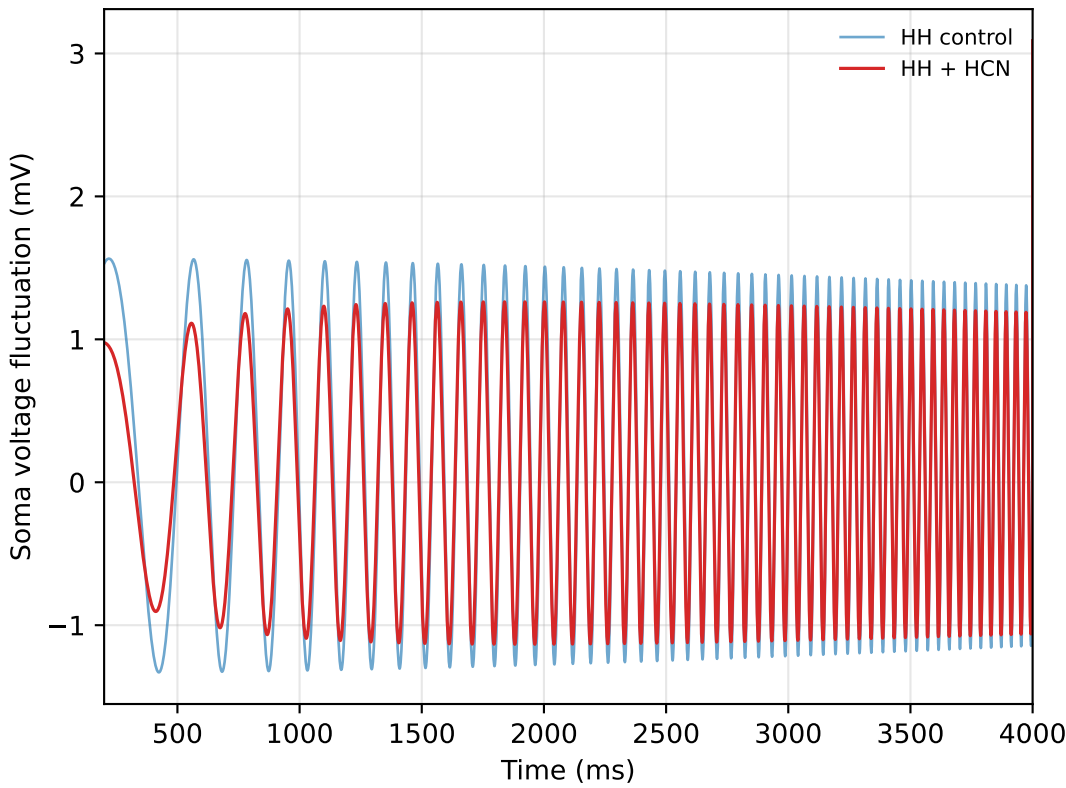


HH + HCN/Ih model: -90 mV subthreshold active resonance

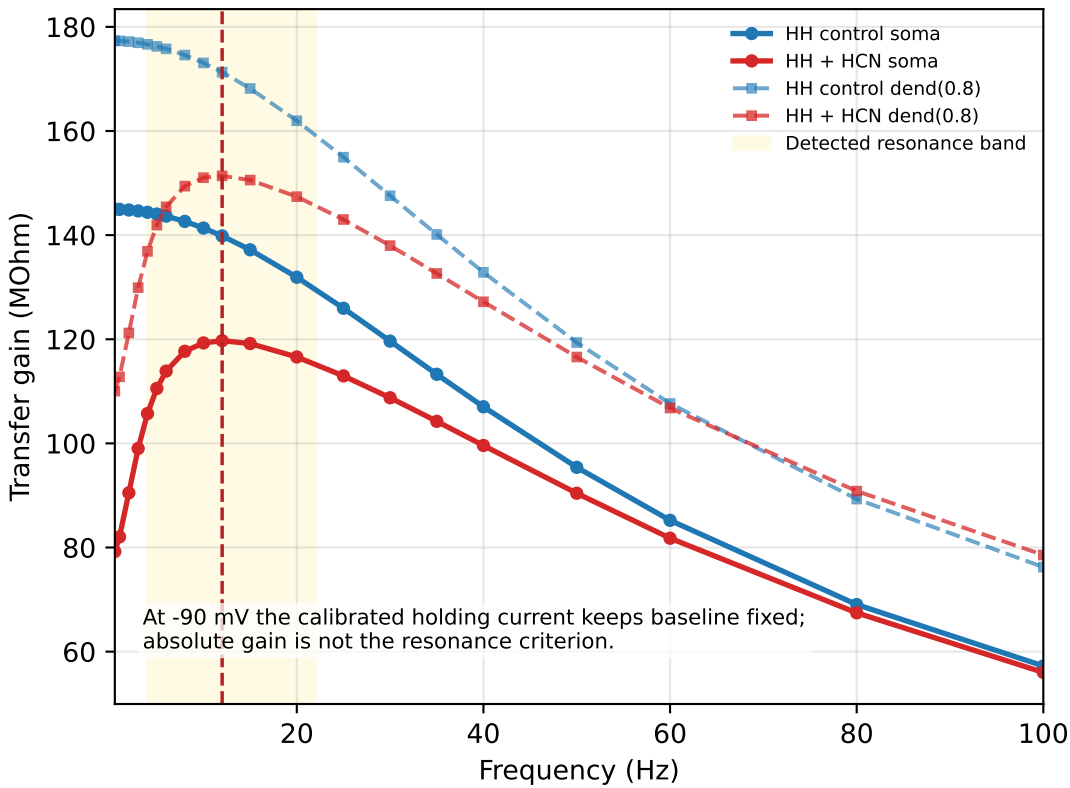
A Hyperpolarization sag with HH background



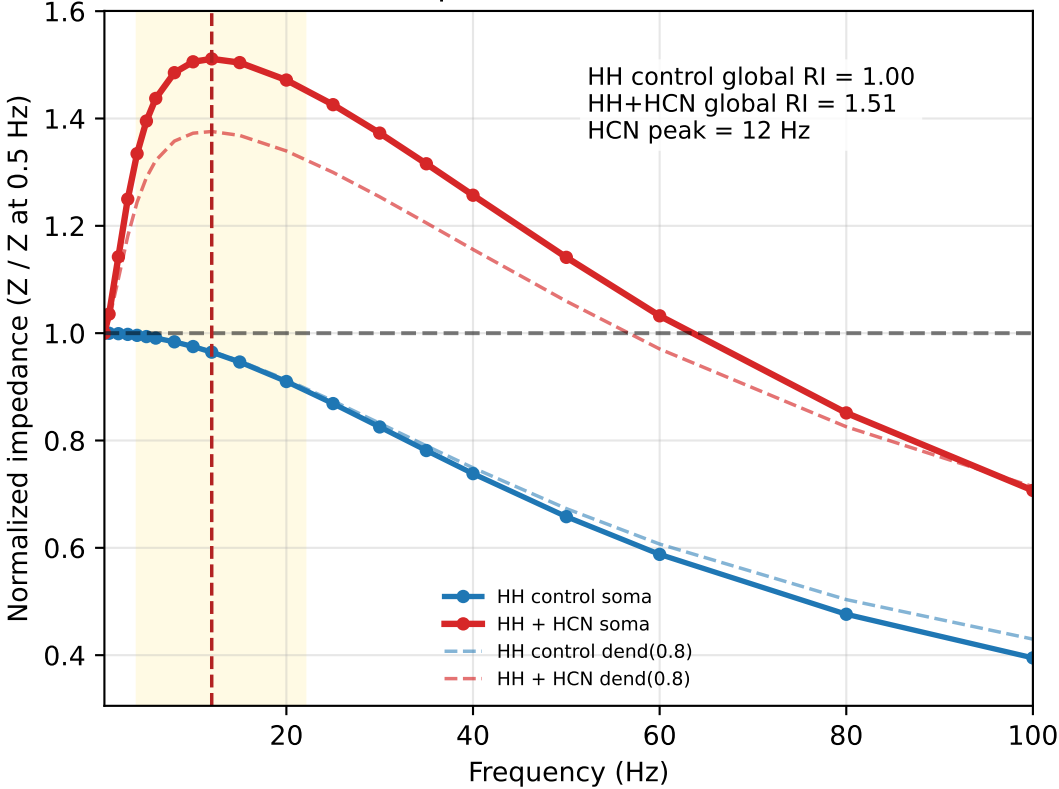
B Subthreshold ZAP trace at -90 mV, no spikes



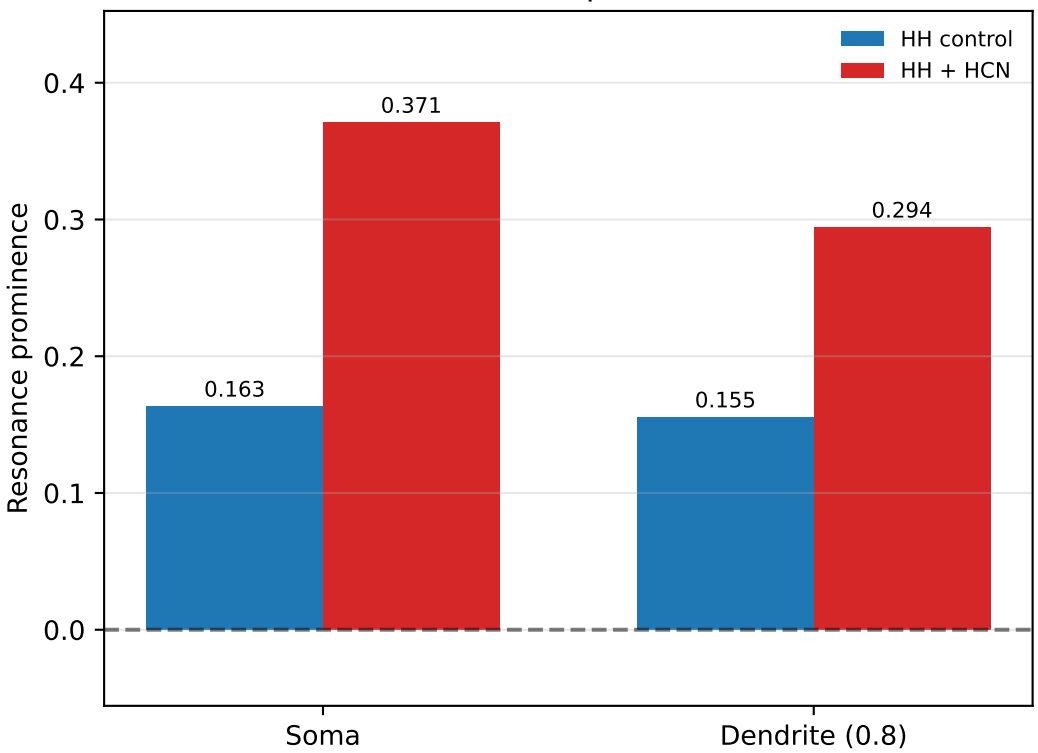
C Absolute transfer gain



D Normalized impedance / active resonance index



E Resonance prominence



Model:

- Soma: pas + hh1 Na/K +/- Ih
- Dendrite: pas +/- Ih
- Input: distal current at dend(0.8)
- Recording: soma and dend(0.8)

Operating point:

- Operating Vm: -90.0 mV
- Holding control: -0.1361 nA -> soma -90.0 mV, dend -84.7 mV
- Holding HCN: -0.1707 nA -> soma -90.0 mV, dend -83.4 mV
- Holding: calibrated current clamp, not voltage clamp
- Fixed g_h : 0.0008 S/cm²
- Soma Ih density: 0.25 x g_h
- AC amplitude: 0.0100 nA
- Frequency range: 0.5-100 Hz
- Resonance band: 4.0-22.0 Hz
- HH effects are expected to be weak at this hyperpolarized Vm.

Validation:

- HH spike validation: True, current=0.08 nA
- ZAP/sweep for resonance: no spikes = True

Results:

- HCN sag recovery: 2.21 mV
- HH control global RI: 1.000
- HH+HCN global RI: 1.511
- HH+HCN global peak: 12.0 Hz
- Theta-band RI: 1.506
- Prominence soma: control=0.163, HCN=0.371
- Prominence dend: control=0.155, HCN=0.294

Conclusion:

HH+HCN global RI > HH control global RI under -90 mV subthreshold conditions. This reflects normalized frequency tuning, not absolute voltage amplification.